Rofhiwa Dagada Data Science exercise 3

① SELECT
product_id,
product_name,
price,

CASE
WHEN price > 1000 THEN 'Expensive'
WHEN price BETWEEN 100 AND 1000 THEN 'mid-range'
WHEN price < 100 THEN 'Budget'

END AS price_category

From products;

product_id	product_name	Price	Price_category
1	Laptop	1200	Expensive
2	Phone	800	mid-range
3	keyboard	45	Budget
4	monitor	300	mid-range
5	mouse	25	Budget

(2) SELECT
order_id,
customer_name,
amount,

CASE
WHEN amount >1000 THEN 'High value'
WHEN amount BETWEEN 500 AND 999,99 THEN 'medium value'
WHEN amount <500 THEN 'Low value'
END AS order_value_category
from Orders;

order_id	customer_name	amount	order_value_category
1	Alice	150.00	Low value
2	Bob	560.00	medium value
3	Charlie	999.99	medium value
4	Diana	45.50	Low value
5	Ethan	1200.00	High value

③ Select
emp_id,
emp_name,
department,
Salary,

CASE
WHEN department = 'IT' AND Salary > 80000 THEN 'Senior IT'
WHEN department = 'HR' AND Salary > 55000 THEN 'Experience HR'
ELSE
'staff'
END AS position_level
from Employees;

Emp_id	Emp_name	Department	Salary	position_level
1	John	IT	85000	Senior IT
2	Sara	HR	60000	Experience HR
3	mark	IT	75000	Staff
4	Lucy	finance	95000	Staff
5	Tom	HR	55000	Staff

(4) SELECT

Student_id,
Student_name,
Score,

WHEN Score > 90 THEN 'A'
WHEN ~~Score~~ Score BETWEEN 80 AND 89 THEN 'B'
WHEN Score BETWEEN 70 AND 79 THEN 'C'
WHEN Score BETWEEN 60 AND 69 THEN 'D'
WHEN Score < 60 THEN 'F'

END AS grade

From Students

Student_id	Student_name	Score	Grade
1	Anna	92	A
2	Ben	76	C
3	Cara	59	F
4	David	83	B
5	Ella	68	D

(5)

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SELECT
    delivery_id,
    delivery_time_minutes,
CASE
    WHEN delivery_time_minutes <= 30 THEN 'fast'
    WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'onTime'
    WHEN delivery_time_minutes > 60 THEN 'Late'
END AS performance
from deliveries;
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delivery_id	delivery_time_minutes	performance
1	45	on-Time
2	80	Late
3	30	fast
4	65	Late
5	100	Late

(6) SELECT
ticket_id,
issue_type,
priority,

CASE
WHEN priority = 3 THEN 'High'
WHEN priority = 2 THEN 'medium'
WHEN priority = 1 THEN 'Low'

END AS priority_Label

from tickets;

ticket_id	issue_type	priority	priority_Label
1	Login issue	1	Low
2	Server down	3	High
3	Slow System	2	medium
4	Email error	2	medium
5	pacsword reset	1	Low

(7) SELECT
Student_id,
days_present,
total_days,
(days_present/total_days) * 100 As attendance_percentage,

CASE
WHEN attendance_percentage >= 90 THEN 'Excellent'
WHEN attendance_percentage BETWEEN 75 AND 100 THEN 'good'
WHEN attendance_percentage <75 THEN 'Needs Improvement'
END AS attendance_Status

from attendance

student_id	days_present	total_days	attendance_percentage	attendance_Status
1	45	50	90	Excellent
2	30	50	60	needs improvement
3	48	50	96	Excellent
4	25	50	50	needs improvement
5	50	50	100	Excellent

(8) SELECT

product_id,
stock_qty,

CASE

WHEN Stock_qty = 0 THEN 'out of stock'
WHEN Stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
WHEN Stock_qty > 5 THEN 'in stock'

END AS stock_Status

From product_inventory

product_id	stock_qty	Stock_Status
1	5	Low Stock
2	0	out of stock
3	25	in stock
4	10	in Stock
5	3	Low Stock

(9) SELECT
Class_Id,
enrolled_Students,

CASE
WHEN enrolled_Students >= 25 THEN 'Large'
WHEN enrolled_students BETWEEN 10 AND 24 THEN 'medium'
WHEN enrolled_students < 10 THEN 'Small'
END AS Class_size_Category
from Classes;

Class_Id	enrolled_Students	Class_Size_Category
1	30	Large
2	25	Large
3	15	medium
4	5	Small
5	20	medium

(10) Select

payment_id,
amount,
payment_method,

CASE
WHEN payment_method = cash AND amount >= 200 THEN 'Eligible for Discount'
Else
NOT Eligible
END AS discount_eligibility
from payments;

payment_id	amount	payment_method	discount_eligibility
1	50	card	NOT eligible
2	200	cash	eligible for discount
3	150	card	Not eligible
4	75	paypal	not eligible
5	300	cash	Eligible for discount